CO1 – ASSESSMENT TOOL 1

IT Application-Based Problem Solving

Scalability Audit of an E-Commerce Platform

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Course code: CSA0608 and Design and analysis of algorithm

1. Aim

To analyze the scalability of an e-commerce platform by testing its performance under different user loads and measuring response time, throughput, and system efficiency to identify potential performance bottlenecks.

2. Requirements

Hardware requirements

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software requirements

- Operating System: Windows 10/11 or Linux
- Programming Language: Python 3.x
- IDE: Visual Studio Code or IDLE
- Libraries: time, random, matplotlib (optional for graphs)

3. Problem Statement

An e-commerce platform must handle multiple users browsing products, searching for items, and placing orders simultaneously. When the number of users increases, the platform may experience longer response times, reduced throughput, and performance degradation.

Therefore, a scalability audit is required to simulate different user loads, evaluate system performance, identify bottlenecks, and determine whether the platform can efficiently handle increasing traffic.

4. Input

The program accepts the number of simulated users as input.

Example:

Enter number of users: 1000

Sample test inputs:

Test case	Number of users
1	100
2	500
3	1,000
4	5,000
5	10,000

5. Algorithm

1. Start the program.
2. Read the number of simulated users.
3. Initialize counters for successful and failed requests.
4. Simulate user requests to the e-commerce platform.
5. Measure the time required to process all requests.
6. Calculate total response time, throughput, and success rate.
7. Display the performance metrics.
8. Repeat the test with different user loads to compare scalability.
9. Stop.

6. Source Code (Python)

```
import time
import random

def scalability_audit(users):
    successful = 0
    failed = 0
    response_times = []
```

```
start_time = time.perf_counter()

for _ in range(users):
    request_start = time.perf_counter()

    # Simulate processing an e-commerce request
    processing_time = random.uniform(0.001, 0.01)
    time.sleep(processing_time)

    if random.random() < 0.98:
        successful += 1
    else:
        failed += 1

    response_times.append(
        time.perf_counter() - request_start
    )

total_time = time.perf_counter() - start_time
avg_response = sum(response_times) / users
throughput = users / total_time
success_rate = (successful / users) * 100

print("\n--- Scalability Audit Report ---")
print("Total Users:", users)
```

```
    print("Successful Requests:", successful)
    print("Failed Requests:", failed)
    print("Average Response Time:",
          round(avg_response * 1000, 2), "ms")
    print("Total Execution Time:",
          round(total_time, 2), "seconds")
    print("Throughput:",
          round(throughput, 2), "requests/second")
    print("Success Rate:",
          round(success_rate, 2), "%")


try:
    users = int(input("Enter number of users: "))

    if users <= 0:
        print("Please enter a positive number.")
    else:
        scalability_audit(users)

except ValueError:
    print("Invalid input. Enter an integer.")
```

output:

```
===== RESTART: C:/Users/aswin/AppData/Local/Programs/Python/Python314/ds.py ====
Enter number of users: 1000

--- Scalability Audit Report ---
Total Users: 1000
Successful Requests: 974
Failed Requests: 26
Average Response Time: 5.81 ms
Total Execution Time: 5.82 seconds
Throughput: 171.9 requests/second
Success Rate: 97.4 %
>
```

7. Result

The program simulates e-commerce requests and generates a scalability audit report containing the total number of users, successful and failed requests, average response time, total execution time, throughput, and success rate.

Expected observation: As the number of simulated users increases, the total execution time generally increases. The program can be executed with different user counts to compare its results**.**

Example report format:

--- Scalability Audit Report ---

Total Users: 100

Successful Requests: 98

Failed Requests: 2

Average Response Time: 5.60 ms

Total Execution Time: 0.60 seconds

Throughput: 166.67 requests/second

Success Rate: 98.00 %

Note: The values above are illustrative, not actual execution results. The program uses randomized processing times and success rates, so results will vary between runs.

8. Limitations

1. Simulated environment: The program does not test a real e-commerce website or server.
2. No concurrent users: Requests are processed sequentially rather than simultaneously.
3. Randomized results: Response times and failures vary between runs.
4. Limited metrics: CPU usage, memory consumption, database performance, and network latency are not measured.
5. No real database: Product searches, payments, and order processing are not connected to an actual database.
6. Limited scalability analysis: Reliable conclusions about real-world performance require load testing with concurrent requests and server monitoring.